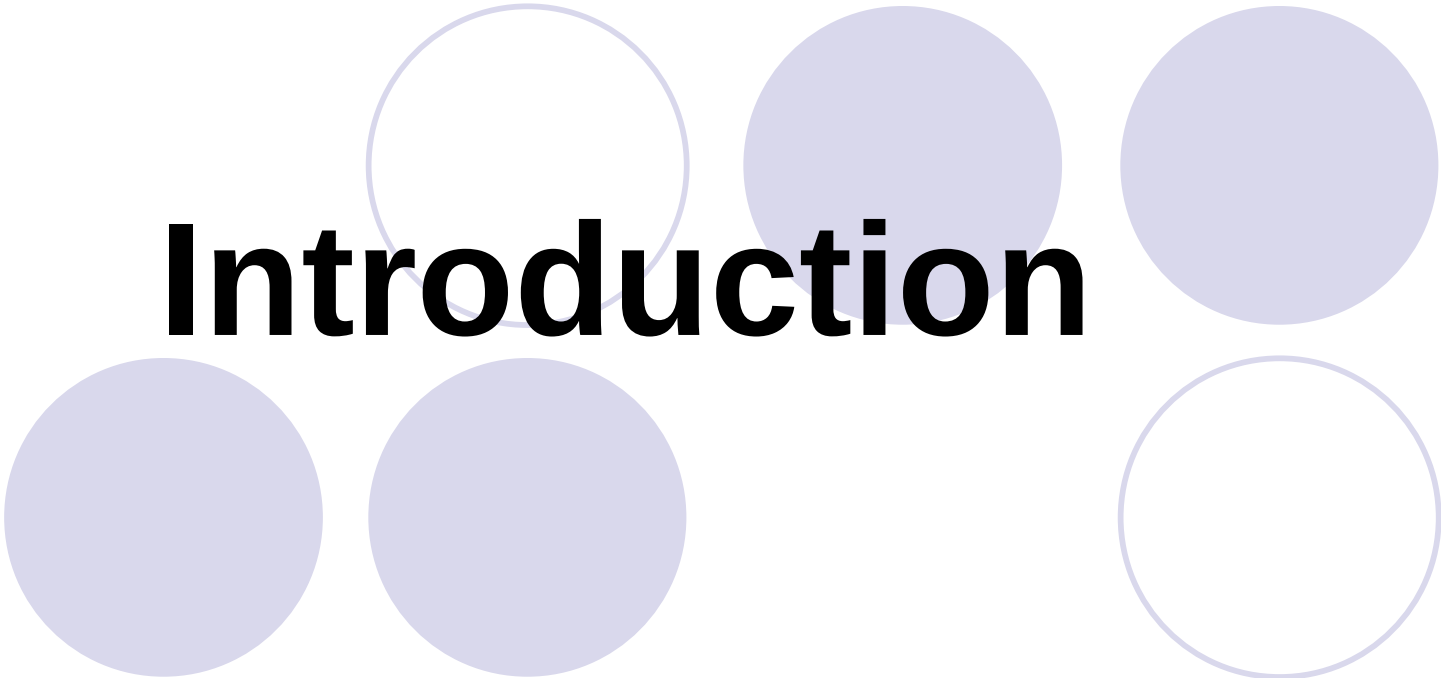


The word "Introduction" is centered on the page. It is surrounded by six circles of varying shades of purple. Two circles are solid, and four are hollow outlines. They are arranged in a loose, abstract pattern around the text.

# Introduction

# Who Am I?

- Dr. Ahsanur Rahman
- Associate Professor, ECE, NSU
- B.Sc. in CSE from BUET
- PhD in CS from Virginia Tech, USA
- Website:

<https://sites.google.com/site/rahmanmahsanur/>

Your name, semester, BSc  
Inst., why this course?

Are you a full-time job-holder?

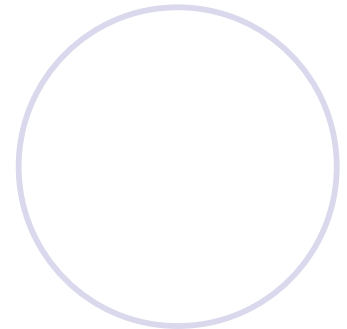
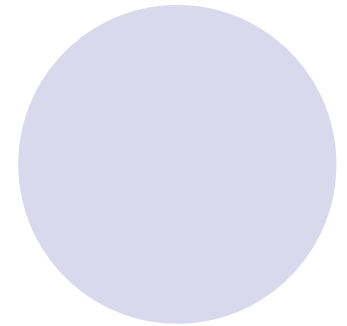
What do you expect from me?

# How to contact me?

- **Best way:** meet me during my office hours
  - **Office:** SAC 914
  - What to do if I am not there?
    - Email me, so that I come ASAP if I am not busy in a meeting
  - Where is my office hours?
    - In front of my door and in Canvas
- **Email:** [ahsanur.rahman@northsouth.edu](mailto:ahsanur.rahman@northsouth.edu)
- **For common requests:**
  - Don't email me
  - Fill up this G-Form: <https://forms.gle/3vMZuUNXKGhp9R1b9>

# Grading system

| Numerical Scores | Letter Grade | Grade Points Per Credit |
|------------------|--------------|-------------------------|
| 93 and above     | A Excellent  | 4.0                     |
| 90 - 92          | A-           | 3.7                     |
| 87 - 89          | B+           | 3.3                     |
| 83 - 86          | B Good       | 3.0                     |
| 80 - 82          | B-           | 2.7                     |
| 77 - 79          | C+           | 2.3                     |
| 73 - 76          | C Average    | 2.0                     |
| 70 - 72          | C-           | 1.7                     |
| 67 - 69          | D+           | 1.3                     |
| 60 - 66          | D Poor       | 1.0                     |
| Below 60         | F* Failure   | 0.0                     |



# Breakdown of Score

- Class participation(CP) – 5%
- Project
  - Proposal Presentation + viva + participation – 10%
  - Initial paper – 10%
  - Project contributions (progress reports, peer review) – 25%
    - Non-writing contributions
  - Final presentation + viva – 15%
  - Final paper – 15%
- Quizzes – 20%

# Ground rules for the exams

- Nobody is allowed to use any electronic device (phone/tab/...) except simple calculator
  - Calculator within mobile phone is not allowed
  - All electronic devices must be kept off and must be kept in the bag during exam
  - Turn off all your alarms in your devices
  - Put bags in front of the exam hall before exam
- **Nobody can go outside the exam-room during the exam. Go beforehand.**
- Nobody is allowed to exchange anything with anybody else

# Ground rules for the exams

- Before I come, prepare+sit in 4 columns which have equal distance between them
- During exam, nobody can talk, move, try to look at others papers – no cheating!
- Question & answer scripts will be distributed to all and a few minutes will be given before the exam starts
  - Write ID, name etc. and read & understand questions during this time
  - The invigilator will ask the students if they have any question/confusion about question, in which case he will explain it. Then the exam starts

# Ground rules for the exams

- Keep totally quiet during the exam
  - No question can be asked
  - No request can be made
- Invigilators have the right to take away anybody's paper and keep it indefinitely as well as can deduct marks if the student adopts any unfair means
- When the invigilator tells the students to stop, everyone must stop writing, check if s/he put down his/her id, name, etc. and submit

# Exam Paper Return Policy

- I return almost all your exam papers (including quiz) in class so that you know your mistakes and your scores. Please write down your score and **return** the paper to me.
- After mid, I send you an Excel sheet mentioning all your scores so far. I keep updating & uploading it later on.
- Please verify that your scores in Excel sheet are indeed correct by matching those with the scores you know from returned papers.
- Don't email me for score related issues, rather fill up this G-form and notify me after class:

<https://forms.gle/3vMZuUNXKGhp9R1b9>

# Class participation (CP)

- Based on the number of classes attended
- Do
  - ask questions
  - answer my questions
    - Earn **bonus** by answering correctly & quickly questions on this class-lecture
      - Bonus marks (0.1 ~ 1) depends on the question complexity
  - Focus on what I am saying
  - Take good preparation
  - If you are late > 20 min, your card swipe won't work; you can get at most 0.5 attendance (manually taken ON THAT DAY)

# Class participation

## ● Don't

- be late in joining the class
- use gadgets or create nuisance
- Email me for attendance correction, Rather:
  - Meet me after class (e.g. when your card isn't working) so that I can correct it right then OR
  - If you missed classes for sickness/other issues, fill up this form: <https://forms.gle/3vMZuUNXKGhp9R1b9>
    - You must upload medical/legal/other documents as evidence to support your case
- Miss classes without any valid reason
  - if you still missed any classes for some valid reasons,
    - Collect class-notes from friends, review those along with slides
    - consult with TA if you don't understand something
  - If you still have problems, come to me with specific questions
    - I can't explain the whole class

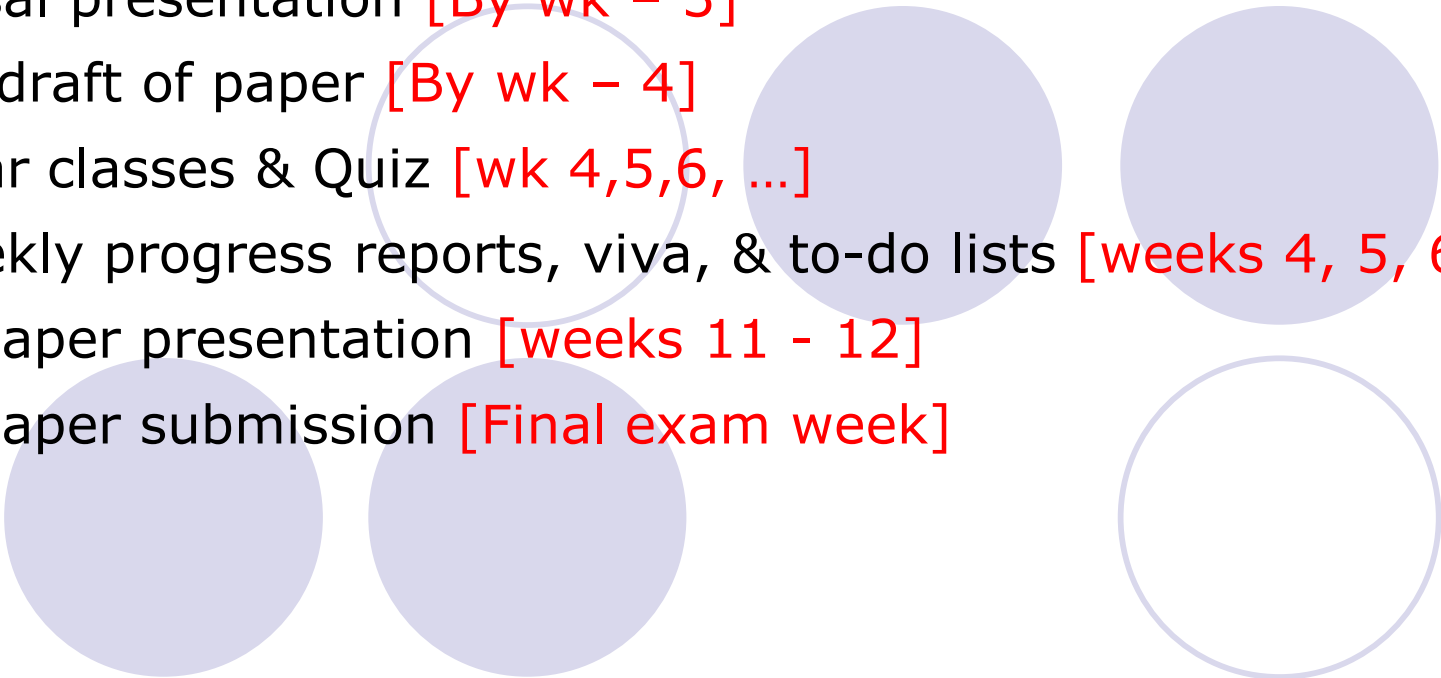
# Quizzes

- 2 quizzes, all will be counted
- Exact schedule will be announced later
- **NO MAKE-UP QUIZZES/mid/final/HW unless there is a real issue (in which case, please contact me right after you come online)!!!**
  - You must provide evidence of your case by filling up [G-Form](#)
  - In such a case, I shall decide how to resolve your case (taking makeup or something else)
  - Please be totally frank with me; I shall do whatever I can to ease your situation
  - But please don't come up with fake excuses to exploit/harass me

# Quizzes

- Do
  - Take good preparation
- Don't
  - copy/cheat
  - delay in submitting

# Evaluation process

- Form group & discuss project ideas with me [By wk-2]
  - Proposal presentation [By wk - 3]
  - Initial draft of paper [By wk - 4]
  - Regular classes & Quiz [wk 4,5,6, ...]
  - Bi-weekly progress reports, viva, & to-do lists [weeks 4, 5, 6,...]
  - Final paper presentation [weeks 11 - 12]
  - Final paper submission [Final exam week]
- 

# Group Discussion/Meeting

- Do
  - offer alternative proposals/solutions
  - Point out strength & weaknesses in offered proposals/solutions
  - come forward to take responsibilities
  - raise questions/confusions if you have any
  - work on your assigned tasks before meeting and discuss the issues/challenges you face
- Don't
  - be absent
  - stay silent
  - cut off others while they are talking

# Individual Grading

- Each team: 2~3 members, depending on project complexity
- Each member must contribute significantly in presenting project, writing, and building project
  - Each will get separate scores
  - A member can fail even if the project is done superbly because s/he didn't contribute enough
- Each member must participate in group discussions
- Evaluation will be based on
  - Individual contribution in data analysis, coding, paper writing, ...
  - Participation of a member during weekly meetings, and during presentations
  - Individual viva during weekly meetings and at the end of semester

# Project Proposal

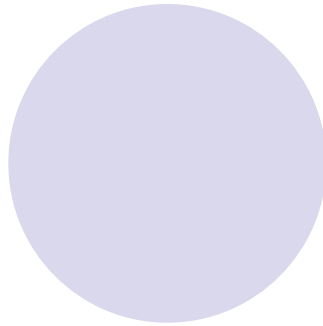
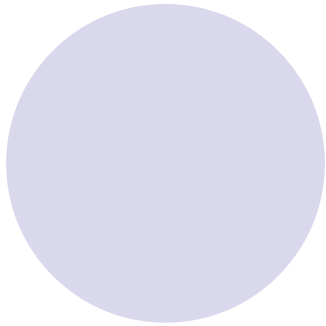
- Discuss with me your ideas by the end of 2nd week to get my feedback (you must get my approval first)
  - Selected problem must be a problem on designing/benchmarking complex/advanced data structures & its operations/algorithms & its applications
    - Specially, data structures for maintaining graphs (static/dynamic) and for range queries
  - Research about the feasibility+usefulness of the problem
- Present the project proposal
- Each group will be given at most 10 min to present & max 5 min for Q&A
- basic concepts relevant to this problem, MOTIVATION: why this problem is important to solve (practical/theoretical benefits) + show that this problem has not been solved before OR can be solved using/adapting existing approaches but that will be inefficient/non-exact, Related Work (details) to show your claims are true, Intuition/high-level ideas to solve your problem and existing solutions, week-by-week plan to complete the project

# Initial paper draft

- Each member must contribute significantly in writing
- All the sections must be completed diligently.
- Copy from internet/friend/anywhere ... will be punished via negative marks.
- Partial copy is still copy...
- write basic concepts, motivation (introduction), Related Work, problem statement, possible/existing solution techniques and week-by-week plan to complete the project
- Evaluation will be based on
  - Writing style
  - Completeness
  - Individual contributions (measured via viva)
  - Correct sentence structure and grammar

# Presentations

- Each member must present some slides
- Prepare yourself for the presentation
- Be formal in dressing; but don't be extravagant
- Give feedback on others' demo & presentations



# Bi-weekly Progress

- Each member must submit individual biweekly report
- Mention the followings in each report
  - Goals of the bi-week
  - Done
  - Doing/Working (partially done but wasn't complete due to error/bug etc.)
- Maintain Google Docs where you write down my feedback and your to-do list